FINAL PROJECT

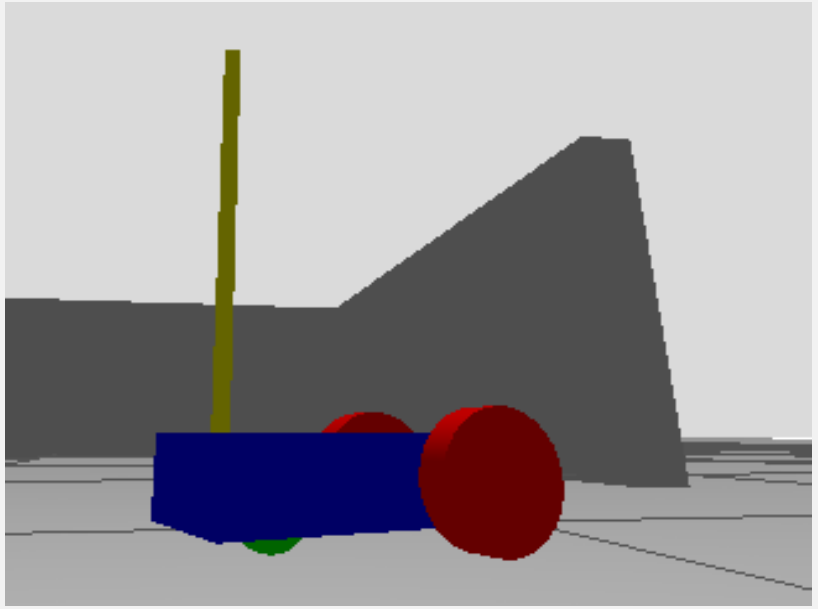
Introduction to Robots

EN.605.613.81.FA25

Logan Griffin

CHANGES TO ROBOT

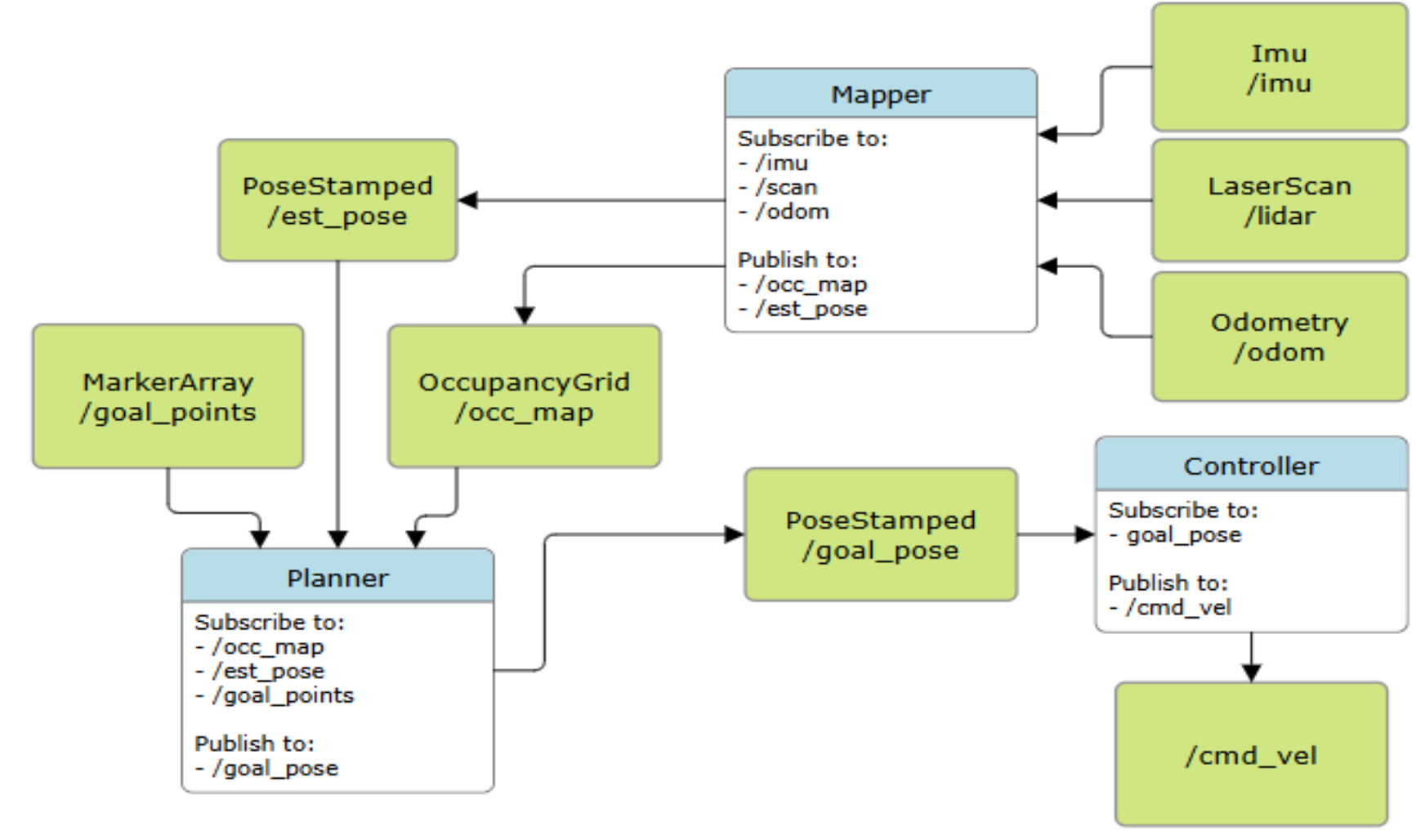
- Moved caster wheel to level robot body
- Added an IMU sensor
- Raised the lidar sensor to avoid hitting back wheels



ALGORITHM

- Mapping
  - EKF
    - Odom as prediction
    - IMU yaw as measurement
  - Occupancy grid with simple probability based on proportion of hits and misses
  - Buffer around occupied cells
- Planning
  - Choose goal location closest to est. pose
  - Grid A* search with Manhattan heuristic
  - Pass goal along path when new estimated pose received

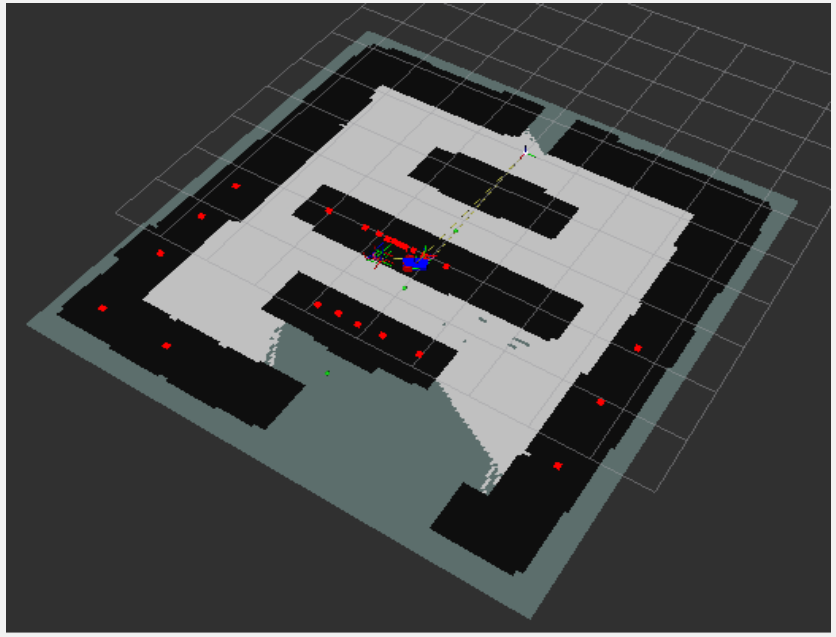
NODE ARCHITECTURE



CHALLENGES

Technical Challenges with ROS

- Resource usage caused inconsistent execution
- Debugging/finding sequence of execution
- Transforms taking time to be available
- Time issue



Algorithm Challenges

- Planning
  - Publishing frequencies caused discrepancies in occupancy grid
  - Robot moving into occupied cell
  - Goal being occupied
  - Choosing appropriate goal from path
- Mapping
  - Occupancy cell probabilities
  - Orientation of occupancy grid
  - Tuning noise for odom and IMU
  - Finding appropriate grid resolution
  - Finding appropriate buffer around obstacles

FUTURE WORK

- Planner
  - Adapt A-star for diagonals
  - Decide how to move when no path is available
  - Choose a goal by path length instead of distance
- Mapper
  - Use circular buffers instead of square to help with slanted walls
  - Reset occupancy grid on intervals to clear noise
  - Adjust number of lidar scans
- Controller
  - Problem
    - calculating error from vehicle's ground truth instead of estimated pose